BIOINFORMATICS INTERNSHIP — CODEALPHA

Task 2: Multiple Sequence Alignment Report

MSA of Human Interleukin Proteins (IL-6, IL-2, IL-4, IL-10, IL-12A)

Tool: Clustal Omega (EMBL-EBI) | Database: UniProt Swiss-Prot

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1. Introduction

Multiple Sequence Alignment (MSA) is a technique used to align three or more biological sequences to identify regions of similarity that reflect shared structure, function or evolutionary history. Conserved regions — the positions that have remained unchanged across sequences — are biologically significant because they suggest those amino acids are critical and cannot be mutated without losing protein function.

For this task, I performed MSA on five human interleukin proteins: IL-6, IL-2, IL-4, IL-10 and IL-12A. I chose this protein family because I had already studied IL-6 in Task 1 using BLAST, so this felt like a natural extension of that work. Interleukins are immune signaling proteins (cytokines) that regulate the immune response. I used Clustal Omega (Sievers et al., 2011) hosted at EMBL-EBI for the alignment, and retrieved all sequences from UniProt Swiss-Prot (UniProt Consortium, 2023).

2. Sequences Selected

All five sequences were downloaded in FASTA format from UniProt Swiss-Prot using their accession numbers. They were combined into one text file and submitted to Clustal Omega:

S.No.	Protein	Full Name	Accession	Key Function
1	IL-6	Interleukin-6	P05231	Immunity, inflammation, hematopoiesis
2	IL-2	Interleukin-2	P60568	T-cell proliferation and immune regulation
3	IL-4	Interleukin-4	P05112	B-cell growth, allergic responses
4	IL-10	Interleukin-10	P22301	Anti-inflammatory cytokine
5	IL-12A	Interleukin-12 subunit alpha	P29459	Links innate and adaptive immunity

Table 1: Five human interleukin proteins used for MSA

Figure 1: UniProt Swiss-Prot entry pages for IL-6 (P05231) and IL-12A (P29459) showing protein details and FASTA sequences used for MSA

After collecting all five sequences, I combined them into a single FASTA file using Notepad. I then opened Clustal Omega at https://www.ebi.ac.uk/jdispatcher/msa/clustalo, selected Protein as the sequence type, pasted all five sequences, set the output format to ClustalW with character counts, and clicked Submit. The results were available within about a minute and included four tabs: Tool Output (raw alignment), Alignments (colored view), Guide Tree, and Phylogenetic Tree — all of which I screenshotted for this report.

Figure 2: FASTA Format Amino Acid Sequences of Human Interleukins (IL-2, IL-4, IL-6, IL-10, and IL-12A)

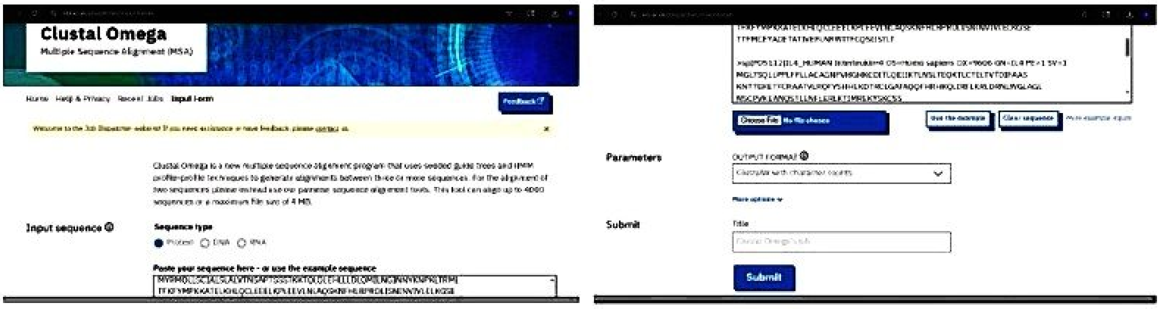

Figure 3: Input and Parameter Setup for Multiple Sequence Alignment of Interleukins in Clustal Omega

4. Results and Analysis

4.1 Alignment Output and Conservation

The alignment output uses three conservation symbols below each aligned block: * (asterisk) means 100% identical amino acid across all five sequences; : (colon) means strongly similar amino acids; and . (dot) means weakly similar amino acids. In my results, there is very limited conservation between these five interleukins — only one fully conserved position (*) was found in the first block, and a few : and . symbols appeared in the third block. This is actually expected because these five proteins, while all interleukins, have evolved to perform very different and specialized immune functions. The few conserved positions that do exist are likely critical for receptor binding or maintaining the protein structure.

The alignment also shows many gap regions (dashes) across all sequences, reflecting the different lengths and sequences of these proteins. IL-6 has 212 amino acids, IL-2 has 153, IL-4 has 153, IL-10 has 178 and IL-12A has 219 — quite different sizes within the same family.

Comparing this to Task 1 is interesting: in Task 1, I compared one protein (IL-6) across multiple species and found 97-100% identity. Here, comparing five different proteins within the same species shows very low conservation — which reflects how functionally specialized each interleukin is.

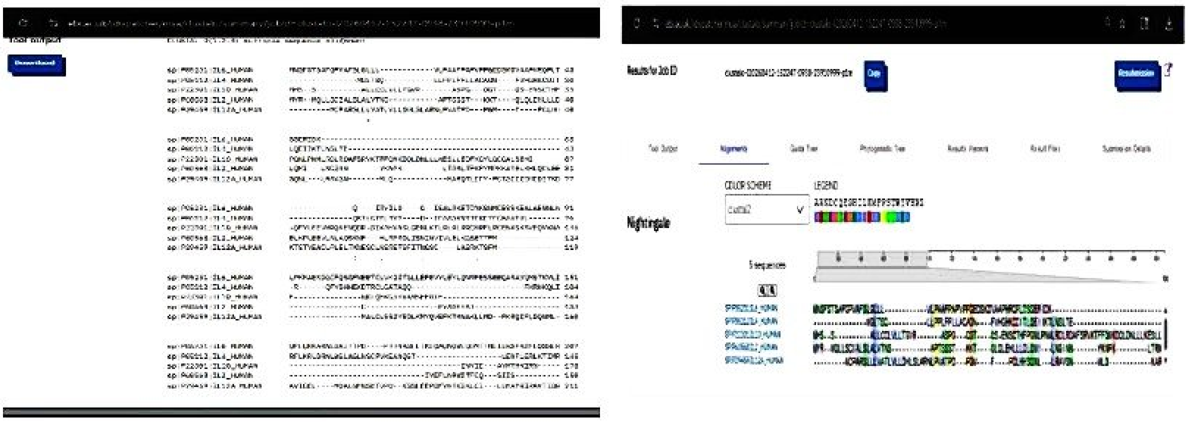

Figure 4: Clustal Omega-Derived Multiple Sequence Alignment and Conserved Region Visualization of Human Interleukin Proteins

4.2 Phylogenetic Tree Analysis

The phylogenetic tree generated by Clustal Omega shows the evolutionary relationships between the five interleukins based on sequence similarity:

Protein	Branch Length	Cluster	Interpretation
IL-6	0.37156	Group 1	Clusters with IL-4 — both involved in B-cell activation
IL-4	0.38399	Group 1	Clusters with IL-6 — share immune differentiation roles
IL-10	0.40779	Separate	Most distinct — unique anti-inflammatory function
IL-2	0.39264	Group 2	Clusters with IL-12A — both regulate T-cell responses
IL-12A	0.34617	Group 2	Shortest branch — most conserved sequence among the five

Table 2: Phylogenetic tree branch lengths and groupings from Clustal Omega

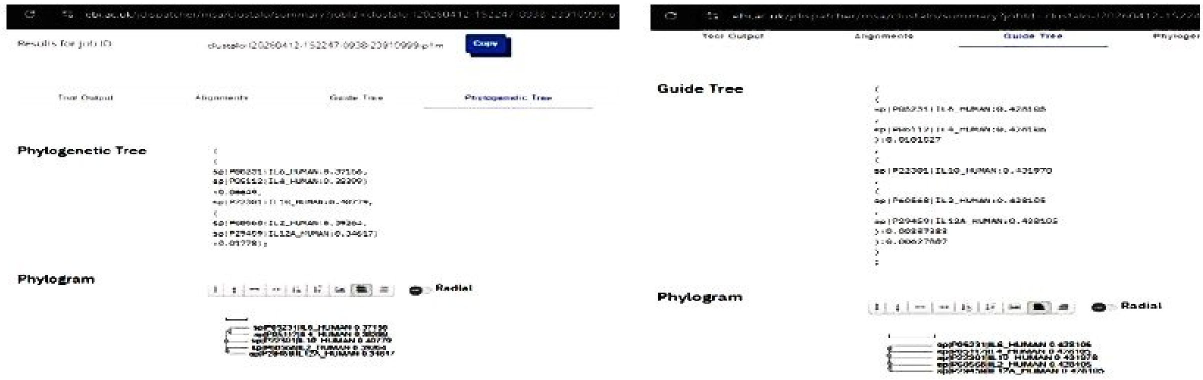

Figure 5: Combined phylogenetic tree and guide tree of human interleukin family proteins (IL6, IL4, IL10, IL2, IL12A)

Two clear clusters emerged from the tree. IL-6 and IL-4 cluster together, which makes sense biologically as both are involved in B-cell activation and immune cell differentiation. IL-2 and IL-12A form a second cluster — both regulate T-cell responses. IL-10 stands alone on a separate branch with the highest branch length (0.40779), reflecting its unique role as an anti-inflammatory cytokine. The Guide Tree confirmed the same groupings, giving me more confidence in these results.

5. Conclusion

This task gave me hands-on experience with multiple sequence alignment using Clustal Omega. The main findings were: the five interleukins show very low overall sequence conservation, reflecting their functional specialization; the few conserved positions found are likely critical for basic protein function; and the phylogenetic tree revealed two main clusters (IL-6/IL-4 and IL-2/IL-12A) that reflect shared biological roles, while IL-10 was the most distinct.

Working on Task 2 after Task 1 was very helpful — studying IL-6 through BLAST first made it much easier to understand its position in the MSA and phylogenetic tree. I will use findings from both tasks when writing my Task 4 report.

6. References

1. Sievers, F., Wilm, A., Dineen, D., Gibson, T.J., Karplus, K., Li, W., Lopez, R., McWilliam, H., Remmert, M., Soding, J., Thompson, J.D., and Higgins, D.G. (2011). Fast, scalable generation of high-quality protein multiple sequence alignments using Clustal Omega. Molecular Systems Biology, 7, 539. https://doi.org/10.1038/msb.2011.75

2. UniProt Consortium. (2023). UniProt: the Universal Protein Knowledgebase in 2023. Nucleic Acids Research, 51(D1), D523-D531. https://doi.org/10.1093/nar/gkac1052

3. Altschul, S.F., Gish, W., Miller, W., Myers, E.W., and Lipman, D.J. (1990). Basic local alignment search tool. Journal of Molecular Biology, 215(3), 403-410. https://doi.org/10.1016/S0022-2836(05)80360-2

4. Tanaka, T., Narazaki, M., and Kishimoto, T. (2014). IL-6 in inflammation, immunity, and disease. Cold Spring Harbor Perspectives in Biology, 6(10), a016295. https://doi.org/10.1101/cshperspect.a016295